Daniel Duclos-Cavalcanti

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Experience

Software Engineer II **Mar 2025 – Present**
Ripple *New York, USA*

- One of **3** engineers building a complete **greenfield** managed-custody product: a **Rust** orchestration layer over Ripple Custody on **AWS Lambda**, **Step Functions**, **EventBridge**, **DynamoDB**, and **KMS** for tier-1/2 bank clients.
- Owned the **actors**, **approvals**, and **domains** APIs: **casbin**-gated maker/checker **quorums**, per-actor **KMS** signing keys bound by **proof-of-possession**, and an **event-sourced**, cursor-paginated stream.
- Built and managed scalable infrastructure using **AWS CDK**: **35 stacks**, ∼1,100 resources per environment, branch-scoped naming allowing concurrent **ephemeral environment** deployments with auto-tear-down.
- Architected an agentic software factory: a spec-driven crew of agents that plan, implement, and merge features semi-autonomously, lifting delivery **∼10×** (**15→155 MRs/mo**, peak **205**) across the product's **∼6-month** build.
- Migrated a **HSM**-backed key **cold-storage** custody vault service from legacy **C#/.NET** to a **Rust** (`axum`) drop-in replacement, serving its batch file-transfer execution API over **gRPC** to two live customers.
- Developed a benchmarking harness in **Rust** (`criterion`) to profile cryptographic operations across the custody's **KMS** backends (**IBM** and **Luna HSMs**, **MPC**), tracking performance regressions in **CI**.

Research Assistant **Jul 2022 – Oct 2022**
TU Munich *Munich, Germany*

- Profiled ML inference across CPUs, GPUs, and Coral Edge TPUs (TensorFlow Lite) via USB traffic analysis (PyShark).
- Established up to 57% of inference time is data transmission to the TPU, informing TensorDSE's hardware mapping.

Embedded Software Engineer – Part Time **Aug 2021 – Jan 2022**
Molabo GmbH *Ottobrunn, Germany*

- Built simulation tooling via Linux virtual CAN, raising GTest coverage 25% on safety-critical motor-controller firmware.
- Owned the team's Jenkins/Docker/CMake **CI** pipeline, supporting 10+ engineers.

Tutor (Embedded Systems Programming Lab) **Apr 2021 – Aug 2021**
TU Munich *Munich, Germany*

- Taught 30+ lab sessions to 12+ students on real-time, multi-threaded FreeRTOS development in C.

Publications

Network Support For Scalable And High Performance Cloud Exchanges | *ACM SIGCOMM '25* **Sept 2025**

- **Onyx**: cloud financial exchange delivering ≤**250 µs** multicast latency, within < 1 µs across 1K receivers.
- **Kernel-bypass** (**DPDK**) sustaining **175K multicast msgs/sec** at **∼50% lower latency** than AWS multicast.
- Cited by researchers and engineers from Jane Street, Google and Microsoft Research.

Projects

Armada: Agentic Software Factory | *Multi-Agent Systems, Rust, SQLite, CLI/Daemon* **2026 – Present**

- A **Rust** daemon (`fleetd`) owns the live fleet behind a single control-plane CLI (`fleetc`), decoupling system and interface, and confining agents' behavior. Actuates on external systems (Jira, GitLab) to drive work from tickets to merged MRs.
- A **SQLite** ledger models the whole system as data, allowing for concurrent access with export, freeze, and rollback; ticking steadily to drive state transitions and run role-specialized agents 24/7.

Viveiro (*M.Sc. Thesis*) | *GCP, ZMQ, Terraform, Python, C++, Distributed Systems* **2024**

- Mitigates **straggler VMs** by probing cluster network performance to build a multicast tree of depth D, fan-out F. Ranks candidate VMs by weighted **p90 one-way delay** and standard deviation from UDP probes.

Technical Skills

Languages: Rust, C++, C, Python, Bash, Golang, Lua
Infrastructure: AWS, GCP, Terraform, CDK, Packer, Docker
Technologies: Linux, FreeRTOS, MPI, OpenMP, gRPC, Protobufs, Tokio
EDA: FPGA, VHDL, Microsemi SmartFusion2 (Libero SoC), Tcl

Education

Technical University of Munich **2024**
M.Sc. Electrical and Computer Engineering *Munich, Germany*

- Visiting Non-Degree Graduate Student: **New York University** – Research & M.Sc. Thesis

Technical University of Munich **2020**
B.Sc. Electrical and Computer Engineering *Munich, Germany*